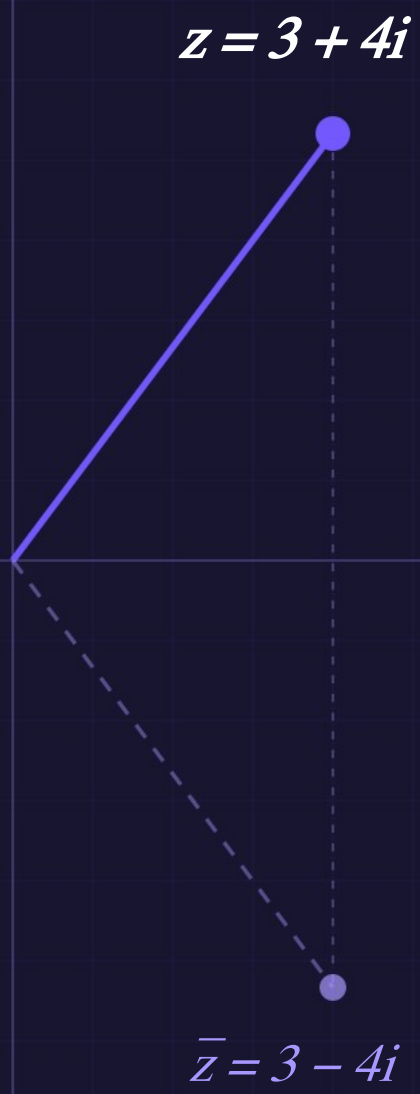
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Salahaddin University-Erbil

School of Science | Department of Information Technology

First-year mathematics



COMPLEX NUMBERS

Properties of Complex Conjugate and Absolute Value

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Why Complex Numbers?

Can we find the square root of a negative number?

With real numbers, no.

The square of a real number is never negative:

$$2^2 = 4 \qquad (-2)^2 = 4 \qquad 0^2 = 0$$

So the equation $x^2 = -1$ has no real solution.

The imaginary unit

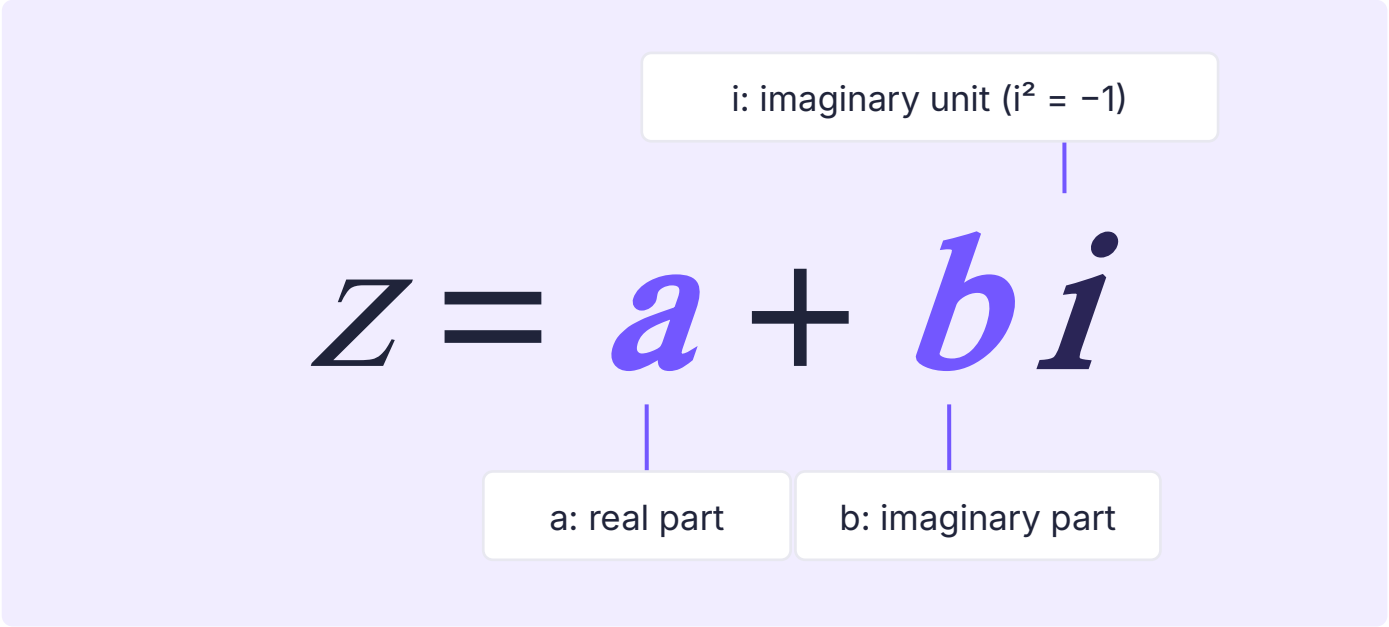
Mathematicians defined a new number i so that:

$$i^2 = -1$$

$$i = \sqrt{(-1)}$$

Now $x^2 = -1$ has two answers:
x = i and x = –i

What Is a Complex Number?



In other words, a complex number is a real number plus an imaginary number.

Examples

$z = 3 + 4i$

real part 3, imaginary part 4

$z = 2 - 5i$

real part 2, imaginary part −5

Understanding Real and Imaginary Parts

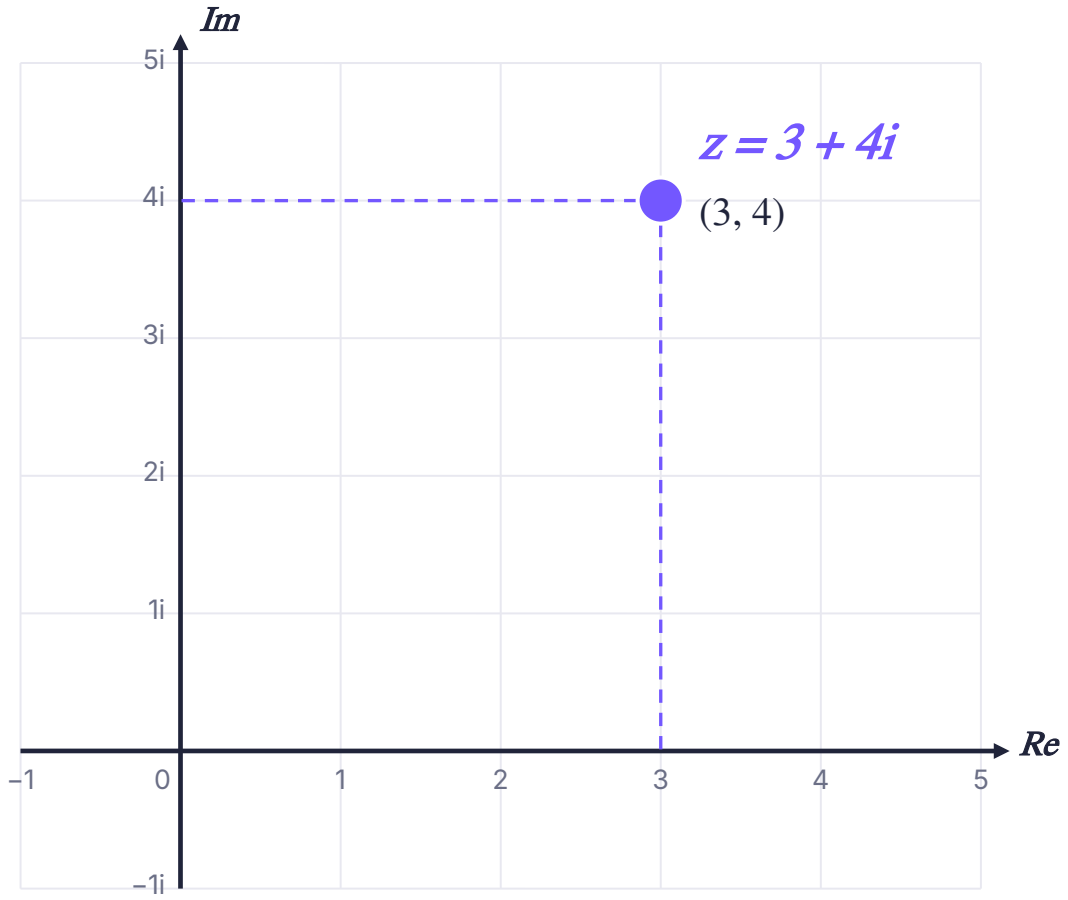
Complex number z	Real part Re(z)	Imaginary part Im(z)
$z = 5 + 2i$	5	2
$z = -3 + 7i$	−3	7

Be careful

The imaginary part is only the number in front of i, without the i. For z = 5 + 2i it is 2, not 2i.

Each part keeps its own sign, so for z = −3 + 7i the real part is −3.

The Complex Plane



Also called the Argand plane.

Horizontal axis: real axis

Shows the real part a.

Vertical axis: imaginary axis

Shows the imaginary part b.

Every complex number is a point

a + bi is drawn at the point (a, b).

To plot 3 + 4i: go 3 to the right, then 4 up.

What Is a Complex Conjugate?

Definition

$$If \quad z = a + bi$$

$$then \quad \bar{z} = a - bi$$

$\bar{z}$ is read "z bar".

How to remember it

Keep the real part.
Change the sign of the imaginary part.

Nothing else changes.

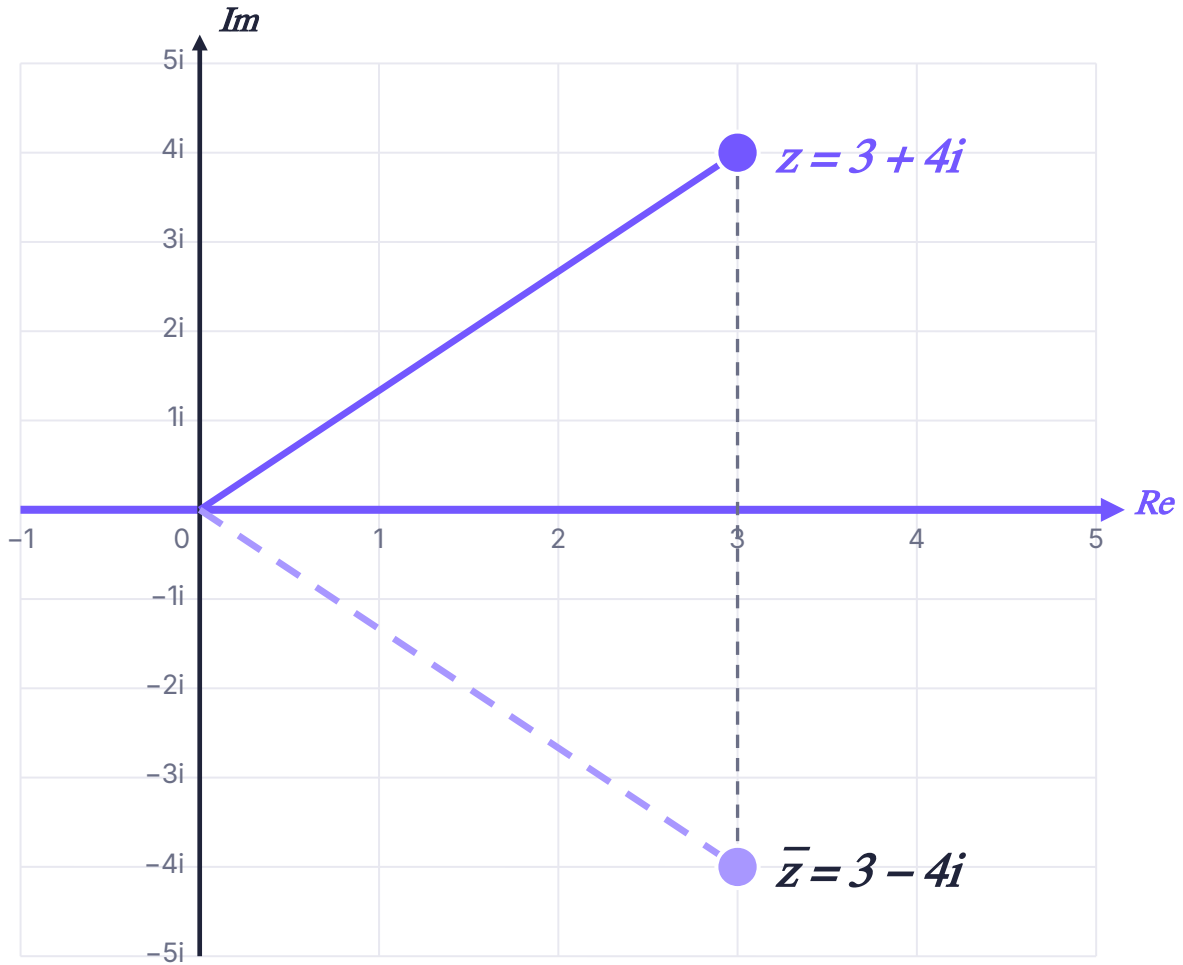
Example

$$z = 3 + 4i$$

$$\bar{z} = 3 - 4i$$

The 3 stays. Only the sign of 4i changes.

Visualizing the Complex Conjugate



Same real part.
Both points are 3 units to the right.

Opposite imaginary part.
4 units up becomes 4 units down.

The real axis acts like a mirror.
The conjugate is the reflection of z across the real axis.

If you fold the graph along the real axis, z lands on $\bar{z}$.

Basic Properties of Complex Conjugates

1. Conjugate of the conjugate

$$\overline{\overline{z}} = z$$

Changing the sign twice gives back the original number.

Example:

$z = 2 + 3i$

$\overline{z} = 2 - 3i$

conjugate again: $2 + 3i$

2. Conjugate of a sum

$$\overline{(z_1 + z_2)} = \overline{z_1} + \overline{z_2}$$

Add first or conjugate first, the result is the same.

Example:

$z_1 = 1 + 2i,\ z_2 = 3 + i$

$z_1 + z_2 = 4 + 3i \ \rightarrow\ 4 - 3i$

$(1 - 2i) + (3 - i) = 4 - 3i$

3. Conjugate of a product

$$\overline{(z_1 z_2)} = \overline{z_1}\ \overline{z_2}$$

Multiply first or conjugate first, the result is the same.

Example:

$z_1 = 1 + i,\ z_2 = 1 + 2i$

$z_1 z_2 = -1 + 3i \ \rightarrow\ -1 - 3i$

$(1 - i)(1 - 2i) = -1 - 3i$

Simple Examples of Complex Conjugates

Example 1

$$z = 6 + 2i$$

$$\bar{z} = 6 - 2i$$

The + in front of 2i becomes −.

Example 2

$$z = -4 - 3i$$

$$\bar{z} = -4 + 3i$$

The real part −4 does not change. Only −3i becomes +3i.

Example 3

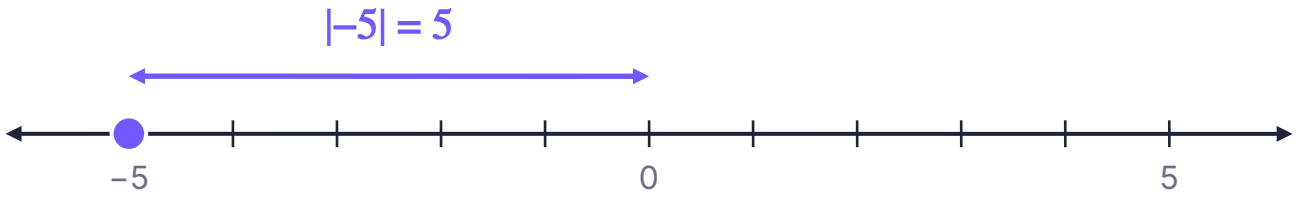
$$z = 7$$

$$\bar{z} = 7$$

There is no imaginary part (b = 0), so nothing changes. A real number is its own conjugate.

Introduction to Absolute Value

For real numbers, absolute value is the distance from 0:



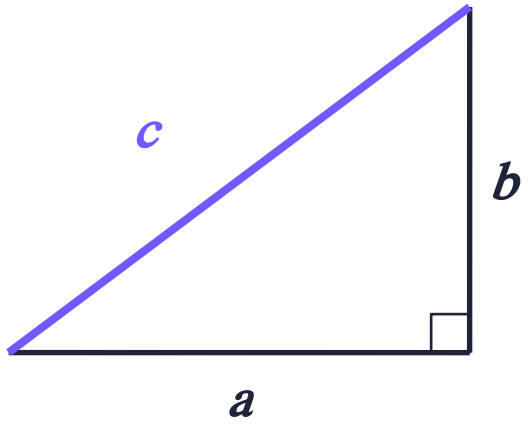
For a complex number, it is the distance from the origin:

If $z = a + bi$,

$$|z| = \sqrt{(a^2 + b^2)}$$

It is also called the modulus of z. We read |z| as "the absolute value of z".

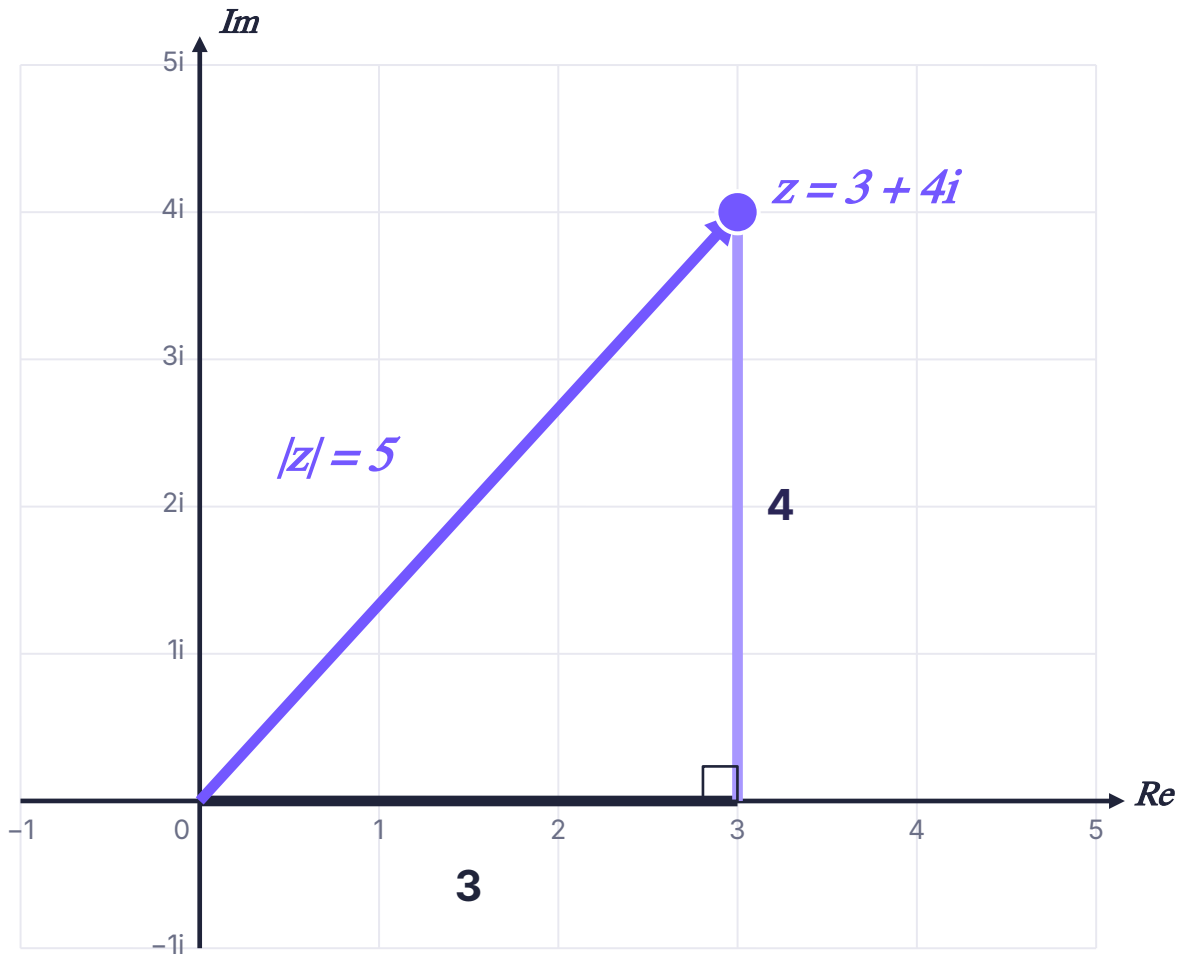
Where does the formula come from?



$$c^2 = a^2 + b^2$$

Pythagoras: |z| is the longest side of a right triangle.

Visualizing Absolute Value



Reading the triangle

- horizontal side: real part = 3
- vertical side: imaginary part = 4
- long side: the distance |z|

$$|z| = \sqrt{(3^2 + 4^2)}$$
$$= \sqrt{(9 + 16)} = \sqrt{25}$$
$$= 5$$

Calculating Absolute Value: Easy Examples

Example 1

$$z = 3 + 4i$$

$$|z| = \sqrt{(3^2 + 4^2)} \quad \text{square each part}$$

$$= \sqrt{(9 + 16)} \quad \text{add}$$

$$= \sqrt{25} \quad \text{square root}$$

$$= 5$$

Example 2

$$z = 5 + 12i$$

$$|z| = \sqrt{(5^2 + 12^2)} \quad \text{square each part}$$

$$= \sqrt{(25 + 144)} \quad \text{add}$$

$$= \sqrt{169} \quad \text{square root}$$

$$= 13$$

Properties of Absolute Value

$|z| \geq 0$

A distance can never be negative.

Example: $|3 + 4i| = 5$

$|z| = 0$ only when $z = 0$

Only the origin is at distance zero.

Example: $|0 + 0i| = 0$

$|\bar{z}| = |z|$

A reflection is the same distance from the origin.

Example: $|3 - 4i| = 5 = |3 + 4i|$

$|z_1 z_2| = |z_1|\,|z_2|$

When we multiply numbers, their sizes multiply.

Example: $|(3 + 4i)\cdot 2| = |6 + 8i| = 10 = 5 \times 2$

The Important Connection

$$z \cdot \bar{z} = |z|^2$$

Why is this useful?

- The result is always a real number (no i).
- It gives $|z|^2$ without a square root.
- We use it for division:

$1 \div (3 + 4i) = (3 - 4i) \div 25$

Check with z = 3 + 4i

$z \cdot \bar{z} = (3 + 4i)(3 - 4i)$

$= 9 - 12i + 12i - 16i^2$

$= 9 - 16i^2$ middle terms cancel

$= 9 + 16$ since $i^2 = -1$

$= 25$

and $|z|^2 = 5^2 = 25$. Same result.

Conjugate vs. Absolute Value

	Complex conjugate $\bar{z}$	Absolute value $\lvert z \rvert$
What it does	Changes the sign of the imaginary part	Gives the size of z
Result	Another complex number	A real number, zero or positive
Formula	$\bar{z} = a - bi$	$\lvert z \rvert = \sqrt{(a^2 + b^2)}$
On the graph	Reflection across the real axis	Distance from the origin
For z = 3 + 4i	$\bar{z} = 3 - 4i$	$\lvert z \rvert = 5$

Real-Life Applications

Electrical engineering

AC voltage and current are written as complex numbers.

Digital signal processing

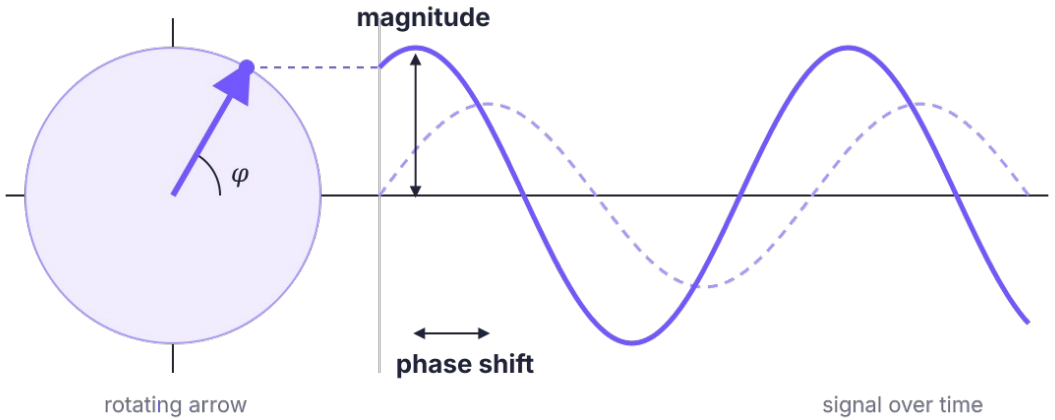
Audio and image filters work with complex values.

Communication systems

Wi-Fi and mobile signals carry data in magnitude and phase.

Sound analysis

Each frequency in a sound has a size and a timing.



Length of the arrow = magnitude, which is $|z|$
Angle of the arrow = phase

So one complex number can store two facts about a signal: its size and its timing.

Practice and Quick Quiz

Try to answer each question first. The answer appears on the next click.

1. Find the conjugate of

$$z = 4 + 6i$$

$$\bar{z} = 4 - 6i$$

2. Find the absolute value of

$$z = 3 + 4i$$

$$|z| = 5$$

3. Find the conjugate of

$$z = -2 - 5i$$

$$\bar{z} = -2 + 5i$$

4. Calculate $|z|$ if

$$z = 5 + 12i$$

$$|z| = 13$$

Conclusion

- A complex number has a real and an imaginary part: $z = a + bi$.
- The conjugate changes the sign of the imaginary part: $\bar{z} = a - bi$.
- The absolute value is the distance from the origin: $|z| = \sqrt{(a^2 + b^2)}$.
- The identity $z\bar{z} = |z|^2$ connects the two ideas.
- Complex numbers are used in mathematics, IT and engineering.

Thank you

Any questions?

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